



Machine Learning Capstone Project Description Document

Name of Project: Intelligent Lending: Building Production-Ready ML & GenAI Models for the Future of Small Business Finance

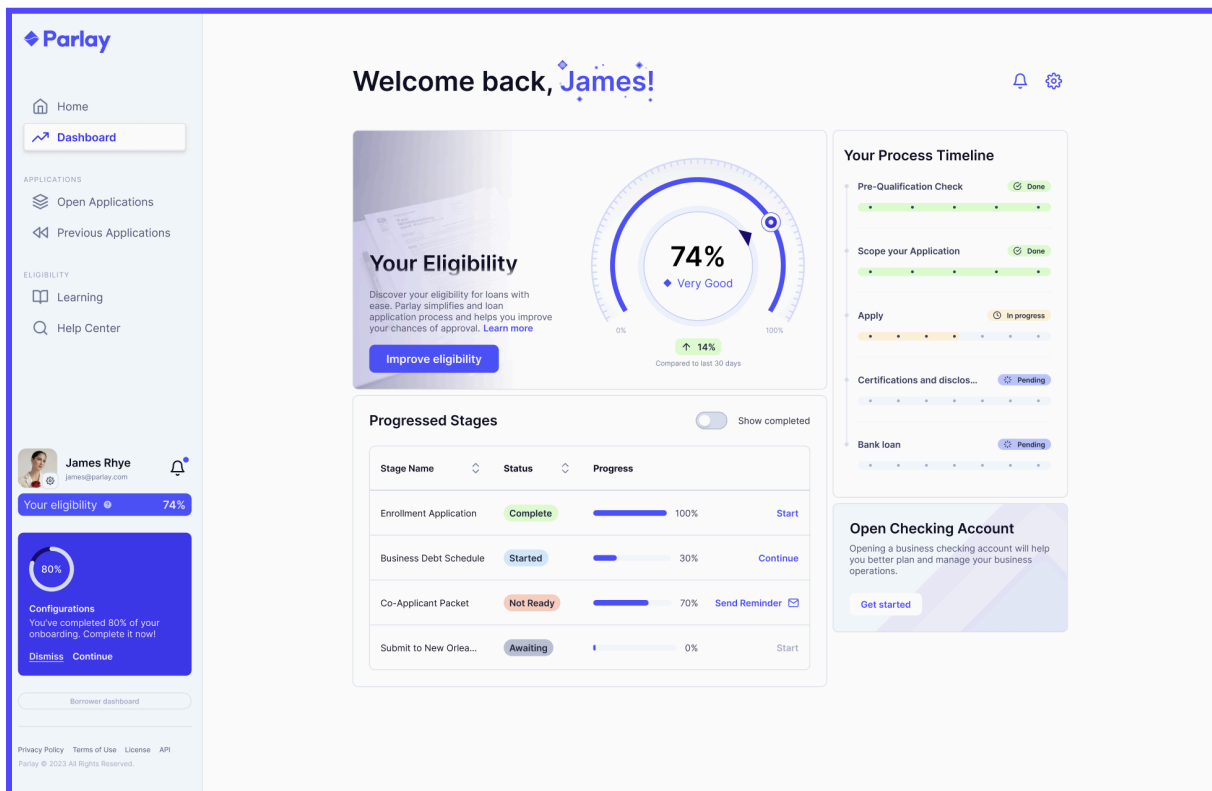
Company Name: Parlay

Mentor Name and Email: James Cho, james.cho@parlay.finance

Part 1: The Project Description

Ideas and Goals:

Parlay Finance is revolutionizing the way financial institutions assess and approve small business loans.. We combine machine learning (ML) techniques and Generative AI (GenAI) models to accurately predict loan eligibility, evaluate fit with lender credit criteria, and generate meaningful insights.



The students in this capstone project will be instrumental in building and refining these ML and GenAI models to enhance our platform's capabilities. Their work will have direct impact by improving loan product recommendations, model explainability, and continuous deployment processes. Parlay operates

in a highly regulated financial environment, so the models must be both high-performing and easily explainable for compliance purposes.

We aim to push the boundaries of classification algorithms (e.g., XGBoost) and LLMs (Large Language Models) to predict not just eligibility, but also why certain products are a good fit. The output from this project will be models ready for production and the deployment of insight-generating LLMs into our real-world platform.

Primary Objectives:

- 1) Deploy Models into Production:
 - Students will focus on getting ML models into production within Parlay's loan recommendation product. This will include developing pipelines and workflows from data preparation to deployment, ensuring models are production-ready.
- 2) Model Development and Benchmarking:
 - Leverage decision trees (e.g., XGBoost) and explore alternative models such as LGBM or Random Forests to ensure we select the best-performing model.
 - Benchmark different ML models to validate which one delivers the highest prediction accuracy and fits within Parlay's business needs.
- 3) LLM Integration and Explainability:
 - Incorporate LLM-based explanations alongside the ML predictions, ensuring model outputs are grounded in feature importance metrics such as SHAP or permutation importance.
 - The LLM will need to generate regulatory-compliant, explainable outputs that can be easily understood by lenders.

Secondary Objectives:

Based on the skill sets, experience, and interest, we are also open to having team(s) work on a subset of our secondary objectives to include:

- 1) Feature Engineering and Data Transformation:
 - Students will explore advanced feature engineering techniques to improve ML model performance.
 - If feature transformations are already present, students will be asked to document and enhance them for model robustness.
- 2) Training and Model Management Documentation:
 - Create comprehensive documentation of the training and evaluation process, including cross-validation/grid search methods.
 - Develop strategies for promoting models to production and tracking model lifecycle management.
- 3) Monitoring and Data Drift Checks:
 - Implement data drift monitoring techniques, such as population stability index (PSI) checks, to ensure input data in production aligns with training data.
- 4) LLM Test Plan Development:
 - Design a test plan for evaluating the LLM's performance, including metrics such as length, comprehensiveness, inclusion of relevant terms, and relevance of explanations.

- Evaluate LLM-generated outputs using both deterministic evaluators and, where appropriate, an LLM-as-judge paradigm for certain metrics.
- 5) AWS Bedrock Benchmarking Pipeline:
- Build a benchmark pipeline on AWS Bedrock to compare the performance of various ML and LLM models under controlled test cases. The goal is to iterate quickly and select the top-performing models for production.

Why This Project Matters:

Working with Parlay Finance provides students with an opportunity to apply advanced data science techniques to real-world challenges in the financial industry. The project emphasizes model accuracy, scalability, and explainability—key priorities in a highly regulated environment. Students will also gain hands-on experience in model deployment and continuous integration practices.

By participating in this capstone, students will:

- Contribute directly to a production-grade product used by financial institutions.
- Work with cutting-edge ML techniques and LLMs to generate actionable insights.
- Enhance technical documentation and model governance for a live platform.
- Learn about regulatory requirements for AI-based solutions in finance.

This project offers real impact and practical experience in deploying data science solutions to address critical problems in financial lending. Join us at Parlay Finance to shape the future of small business lending!

Structure:

Week 1: Introduction and Team Formation

Week 2-3: Current data model onboarding and ideation

Week 4-10: Weekly sprint outcome defined and executed

Week 11-12: Deliverable Handoff and Presentation preparation

Challenges:

Small business lending isn't as straightforward as it should be, it may take time for students to settle in and get comfortable with terms and processes. But don't worry too much, you will have access to all relevant research, products, and our team to ensure that you can get up to speed!

Expectations:

We prefer self driven students that work autonomously and will support a work plan and reporting cadence that the team creates. Students are expected to make progress each week, engaging actively in team meetings and contributing to the project's development. While we estimate the work will be 8-10 hours a week per student, we prioritize outcomes over output. If the team can achieve the desired outcome each week in less time that is acceptable.

Team Dynamics and Leadership:

Teams will self-organize with elected leaders, and mentors will provide as much involvement as needed, offering daily feedback and support if required. The students are not obligated to communicate with Parlay daily, daily communication is offered to ensure students are not blocked. We will invite team students to our company Slack to communicate. We ask that students provide a work plan and share updates as they feel necessary to brief out progress against the plan.

“Given” Elements: We have an existing data model the team(s) can use as a starting point. Additional resources are covered below.

Resources:

- Access to Parlay’s historical data and research
- Collaboration with the Parlay team, including access to domain experts.
- Guidance and mentorship from Parlay's engineers.
- Access to computing resources and tools required for model development and testing.

Specifications and Expectations:**Part 1:** Introduction and Team Formation (Weeks 1)

- Objective: Introduction to the project, Parlay tea, and available resources
- Expectations:
 - Students familiarize themselves with the data's potential.
 - Formation of teams and election of team leaders.
 - Initial mentor meeting to set project goals and expectations.

Part 2: Data Model onboarding and ideation (Weeks 2-3)

- Objective: Detailed brainstorming and conceptualizing potential use cases.
- Expectations:
 - In-depth discussions and creative sessions.
 - Selection of a few promising ideas for further exploration.

Part 3: Weekly Sprints (Weeks 4-10)

- Objective: Thorough development and analysis of the chosen use case.
- Expectations:
 - Detailed feasibility study including data analysis, technical considerations, and potential impact.
 - Begin initial prototyping or model development.

Part 4: Deliverable Handoff and Presentation Preparation (Weeks 11-12)

- **Objective:** Finalize the project, prepare for the final presentation, and compose documentation as needed.
- **Expectations:**
 - Completion of prototype or model development.
 - Preparation of a comprehensive presentation covering all project aspects.
 - Writing documentation so the data model can be used by other engineers.

Deliverables:

At the end of this project, each team is expected to submit **two key deliverables**.

- The first deliverable is a **fully documented ML/LLM pipeline with models ready for production**, comprehensive documentation of the feature engineering, training, and evaluation processes, as well as test plans and benchmarks for monitoring model performance and explainability.
- The second is a **presentation**, including the brainstorming process, selection and development of the chosen use case, insights, challenges, and overall learnings.

Part 2: Student and Team Options

- 1) **Number of Teams you prefer for this Project:** We are open to two to three teams if there is enough interest from the students.
- 2) **Number of students in a Team:** Ideally five, no less than three
- 3) **Meetings**
 - a. All Mentors will meet with their Team(s) via Zoom around Week 3 (~ Feb 2 – Feb 9) to “kick-off” the Project.
 - b. **Meeting per month:** 2-4 and it is dependent on the students’ preference.
 - c. **Asynch check-ins:** *At most* we would have 1 check-in/meeting per week either on a video call or asynchronously using Slack
- 4) **What “style” of leadership do you expect from your team(s)?**

Medium: Teams will elect their own Team Leader and will mostly but not fully lead, run, and manage the project. Mentor participation will be “medium”, such as weekly feedback, answering questions, medium direction. In this case, there are some specific steps and expectations. The Team AND the Mentor together will make the Project design and direction decisions. We will be available to help and provide feedback but ultimately the design decision for the project will mainly rest on the team.
- 5) **What is the nature of the project?**

The project is very open-ended and research based. We are operating on the level of **ideas/concepts** for this project. This project will consist of two main parts.

- The first part will be a **divergent phase** where the students get onboarded to working with telematics data and we try to think of the best way to imagine as many use cases with the data and external augmented data we can.
- Then the second part of the project will be selecting one of the use cases and developing it further.

6) **Do students need to be US Citizens to participate in the project?** No.

7) **What backgrounds or prerequisites would you prefer that students have?** Diversity of background and experience is welcomed! We are a team of US immigrants and veterans who care deeply about helping underserved communities. The following are preferred but not required qualifications:

- Python
- Machine Learning (Reinforcement learning a bonus)
- REST APIs
- Building or refining data models
- Working/integrating with web applications
- Former small business owner
- Finance industry experience

8) **How many hours per week do you expect from each student on your team?** 8-10 hours. However, we prioritize outcomes over output. If the team can achieve the desired outcome each week in less time that is acceptable.

9) **Please include here any links or data or websites that might help students to better understand the overall nature of the focus of the project and our team.**

- Company Website: <https://parlay.finance/>
- Founder LinkedIns
 - CEO - Alexandra McLeod: <https://www.linkedin.com/in/mcleodalexandra/>
 - COO - Jay Long: <https://www.linkedin.com/in/jayllong/>
 - CTO - James Cho: <https://www.linkedin.com/in/jcho-parlay/>
- Founder Individual User Manuals
 - Alex - <https://app.joyn.one/w/alexmcleod>
 - Jay - <https://app.joyn.one/w/jaylong>
 - James - <https://app.joyn.one/w/jamescho>

10) **In one paragraph, please describe the expectations you have of your student team with respect to the project, deliverables, meetings, engagement, etc.**

- We expect our student team to approach the project with professionalism, curiosity, and collaboration. Students should actively engage in regular progress meetings (weekly or biweekly) to ensure alignment with project milestones and provide status updates. Deliverables will include a fully documented ML/LLM pipeline with models ready for

production, comprehensive documentation of the feature engineering, training, and evaluation processes, as well as test plans and benchmarks for monitoring model performance and explainability. We expect students to be proactive in problem-solving, communicate challenges early, and be open to feedback and iteration. Engagement with our team will be crucial, and we encourage a spirit of ownership—treating the project as if they were deploying it in a real-world environment.